

🎓 教育背景

清华大学

◦ 工学博士，计算机科学与技术。导师：王智教授。

中国，深圳

2024.08 – 至今

清华大学

◦ 工程硕士，计算机技术。导师：王智教授。

中国，深圳

2020.06 – 2023.08

北京师范大学

◦ 理学学士，计算机科学与技术。

中国，北京

2016.08 – 2020.06

🏢 研究与工作经历

卡内基梅隆大学，计算机学院，美国

2026.04 – 2026.10

◦ 访问研究助理，合作导师：Minchen Li 教授；研究方向：Real2Sim 与机器人仿真。

◦ 主导 **CoDimRecon**：从多视角 RGB 重建包含刚体、关节物体及可变形物体的可编辑场景；以几何先验约束尺度与布局，以生成网格为参考，为关节物体显式建模关节。

◦ 通过分类 Agent 工作流重建中心线与半径、带厚度的流形壳体及可供体网格剖分的水密实体；利用可复用仿真技能初始化物理模型，通过机器人仿真测试修正运动、几何、数值与材料问题，展示可保留折痕的塑性纸张折叠。

ABB 机器人拾取流水线，数据采集与策略微调

2025.11 – 2025.12

◦ 使用 **LeRobot** 搭建双摄像头数据采集管线，基于采集数据微调 $\pi_{0.5}$ ，用于拾取流水线上的动态物体。

清华大学，高效三维表示与压缩

2023.10 – 2025.10

◦ 开发 **MesonGS**、**SizeGS**、**MesonGS++**，融合 RAHT/八叉树、混合精度量化、混合整数优化与 CUDA 并行搜索；超参数搜索最多加速 $8\times$ ，压缩率超过 $34\times$ ，并保持渲染保真度。

清华大学，Armol / SkyML：强化学习

2020.07 – 2023.08

◦ 面向多云推理服务选择构建组合 RL，设计图像特征状态、服务子集动作与精度/成本奖励；基于 **SAC**、经验回放和双 Q 网络训练，与 **PPO/TD3** 对比，在相近精度下将费用降低 **67%**。一作发表于 **INFOCOM 2022 / TMM 2024**；代码已开源。

腾讯，优图实验室，上海

2022.01 – 2022.06

◦ 模型加速实习生，使用 **Faster-RCNN**、**ONNX**、**OpenVINO** 与 **TensorRT**。

📄 论文与预印本 (* 共同一作；本人姓名加粗)

- [1] **CoDimRecon: Agentic Reconstruction of Sim-Ready 3D Scenes with Deformable Curves, Surfaces, and Volumes**
Shuzhao Xie*, Lelin Wang*, Guying Lin, Zhi Wang, Minchen Li. *arXiv preprint*, 2026.
- [2] **MesonGS++: Post-training Compression of 3D Gaussian Splatting with Hyperparameter Searching**
Shuzhao Xie*, Junchen Ge*, Weixiang Zhang, Jiahang Liu, Chen Tang, *et al.* *arXiv preprint*, 2026.
- [3] **Tail-Aware Post-Training Quantization for 3D Geometry Models**
Sicheng Pan*, Chen Tang*, **Shuzhao Xie***, Ke Yang, Weixiang Zhang, *et al.* *arXiv preprint*, 2026.
- [4] **SizeGS: Size-aware Compression of 3D Gaussian Splatting via Mixed Integer Programming**
Shuzhao Xie*, Jiahang Liu*, Weixiang Zhang, Shijia Ge, Sicheng Pan, *et al.* *ACM MM*, 2025. *Best Paper Candidate*.
- [5] **SD-GS: Structured Deformable 3D Gaussians for Efficient Dynamic Scene Reconstruction**
Wei Yao*, **Shuzhao Xie***, Letian Li, Weixiang Zhang, Zhixin Lai, *et al.* *arXiv preprint*, 2025.
- [6] **MesonGS: Post-training Compression of 3D Gaussians via Efficient Attribute Transformation**
Shuzhao Xie, Weixiang Zhang, Chen Tang, Yunpeng Bai, Rongwei Lu, *et al.* *ECCV*, 2024.
- [7] **SkyML: A MLaaS Federation Design for Multicloud-based Multimedia Analytics**
Shuzhao Xie, Yuan Xue, Yifei Zhu, Zhi Wang. *TMM*, 2024.
- [8] **Cost Effective MLaaS Federation: A Combinatorial Reinforcement Learning Approach**
Shuzhao Xie, Yuan Xue, Yifei Zhu, Zhi Wang. *INFOCOM*, 2022.
- [9] **IGen: Scalable Data Generation for Robot Learning from Open-World Images**
Chenghao Gu*, Haolan Kang*, Junchao Lin*, Jinghe Wang, Duo Wu, **Shuzhao Xie**, *et al.* *CVPR*, 2026.
- [10] **Splatwizard: A Benchmark Toolkit for 3D Gaussian Splatting Compression**
Xiang Liu, Yimin Zhou, Jinxiang Wang, Yujun Huang, **Shuzhao Xie**, *et al.* *CVPR Findings*, 2026.
- [11] **Treat Domain-Specific Languages as Design Variables in LLM Agents**
Letian Li, Wenyan Jiang, Xin Yang, **Shuzhao Xie**, Chenghao Gu, *et al.* *NeurIPS, Position Paper Track*, 2026.
- [12] **HDSL: A Hierarchical Domain-Specific Language for Structured 3D Indoor Scene Generation and Localized Editing with LLM Agents**
Letian Li, Chao Shen, **Shuzhao Xie**, Chenghao Gu, Zhengxiao He, *et al.* *Asia ACL*, 2026.
- [13] **GeniWorld: A Generalizable Interactive World Model for Robotic Manipulation via Visual Actions**
Chenghao Gu, Hanyang Yu, Jingbo Zhang, Haitao Lin, Wenya Zhang, Jinghe Wang, Hanglei Jin, **Shuzhao Xie**, *et al.* *arXiv preprint*, 2026.
- [14] **RoboStream: Weaving Spatio-Temporal Reasoning with Memory in Vision-Language Models for Robotics**
Yuzhi Huang*, Jie Wu*, Weijue Bu*, Ziyi Xiong, Gaoyang Jiang, Ye Li, Kangye Ji, **Shuzhao Xie**, *et al.* *ECCV*, 2026.
- [15] **CurvSpec: Adaptive Multi-Curvature Learning for Partial Relevant Video Retrieval**
Zhen Liu, Letian Li, Jinpeng Wang, **Shuzhao Xie**, Yuzhi Huang, *et al.* *ACM MM*, 2026.

- [16] **TAU-Bench: From Anomaly Instance Tracking to Fine-Grained Video Anomaly Understanding**
Kepeng Yang*, Dongxuan Liu*, Rongxin Gao*, Zixin Su, Rui Wu, Shuzhao Xie, et al.. arXiv preprint, 2026.
- [17] **DynTrace: Tracking Dynamic Object Evidence for 4D Spatio-Temporal Reasoning in MLLMs**
Rongxin Gao, Yuzhi Huang, Dongxuan Liu, Chu Li, Zhenye Wang, Jie Wu, Shuzhao Xie, et al.. arXiv preprint, 2026.
- [18] **Understanding Bias Terms in Neural Representations**
Weixiang Zhang, Boxi Li, Shuzhao Xie, Chengwei Ren, Yuan Xue, et al.. NeurIPS, 2025.
- [19] **Music-Aligned Holistic 3D Dance Generation via Hierarchical Motion Modeling**
Xiaojie Li, Ronghui Li, Shukai Fang, Shuzhao Xie, Xiaoyang Guo, et al.. ICCV, 2025.
- [20] **EVOS: Efficient Implicit Neural Training via EVolutionary Selector**
Weixiang Zhang, Shuzhao Xie, Chengwei Ren, Siyi Xie, Chen Tang, et al.. CVPR, 2025.
- [21] **Enhancing Implicit Neural Representations via Symmetric Power Transformation**
Weixiang Zhang, Shuzhao Xie, Chengwei Ren, Shijia Ge, Mingzi Wang, et al.. AAAI, 2025.
- [22] **Expansive Supervision for Neural Radiance Fields**
Weixiang Zhang, Wei Yao, Shuzhao Xie, Shijia Ge, Chen Tang, et al.. ICME, 2025.
- [23] **Lungmix: A Mixup-Based Strategy for Generalization in Respiratory Sound Classification**
Shijia Ge, Weixiang Zhang, Shuzhao Xie, Baixu Yan, Zhi Wang. ICASSP, 2025.
- [24] **PulmoScan: A Practical Pulmonary Disease Pre-Screening System**
Baixu Yan, Shijia Ge, Meizi Lu, Weixiang Zhang, Shuzhao Xie, et al.. ICASSP, 2025.
- [25] **TextIR: A Simple Framework for Text-based Editable Image Restoration**
Yunpeng Bai, Cairong Wang, Shuzhao Xie, Chao Dong, Chun Yuan, et al.. TVCG, 2025.
- [26] **VGGT-DP: Generalizable Robot Control via Vision Foundation Models**
Shijia Ge, Yijun Liu, Yinxin Zhang, Shuzhao Xie, Weixiang Zhang, et al.. arXiv preprint, 2025.
- [27] **RFQuant: Retraining-free Model Quantization via One-Shot Weight-Coupling Learning**
Chen Tang*, Yuan Meng*, Jiacheng Jiang, Shuzhao Xie, Rongwei Lu, et al.. CVPR, 2024.
- [28] **A Joint Approach to Local Updating and Gradient Compression for Efficient Asynchronous Federated Learning**
Jiajun Song, Jiajun Luo, Rongwei Lu, Shuzhao Xie, Bin Chen, et al.. Euro-Par, 2024.
- [29] **WATER-GS: Toward Copyright Protection for 3D Gaussian Splatting via Universal Watermarking**
Yuqi Tan, Xiang Liu, Shuzhao Xie, Bin Chen, Shu-Tao Xia, et al.. arXiv preprint, 2024.
- [30] **DragScene: Interactive 3D Scene Editing with Single-view Drag Instructions**
Chenghao Gu, Zhenzhe Li, Zhengqi Zhang, Yunpeng Bai, Shuzhao Xie, et al.. arXiv preprint, 2024.
- [31] **Tuning-Free Visual Customization via View Iterative Self-Attention Control**
Xiaojie Li, Chenghao Gu, Shuzhao Xie, Yunpeng Bai, Weixiang Zhang, et al.. arXiv preprint, 2024.

🏆 荣誉与奖项

- ACM Multimedia 最佳论文候选（前 0.5%）。 2025
- ACM ICPC/CCPC 区域赛铜牌（×2）。 2018, 2019

🔧 核心能力

- **Real2Sim / RL**: 多视角重建、可变形仿真、SAC、状态/动作/奖励设计、PPO/TD3 对比评测。
- **工程能力**: Python、C/C++、CUDA、Bash、PyTorch; LeRobot、Blender、UIPC、Genesis。